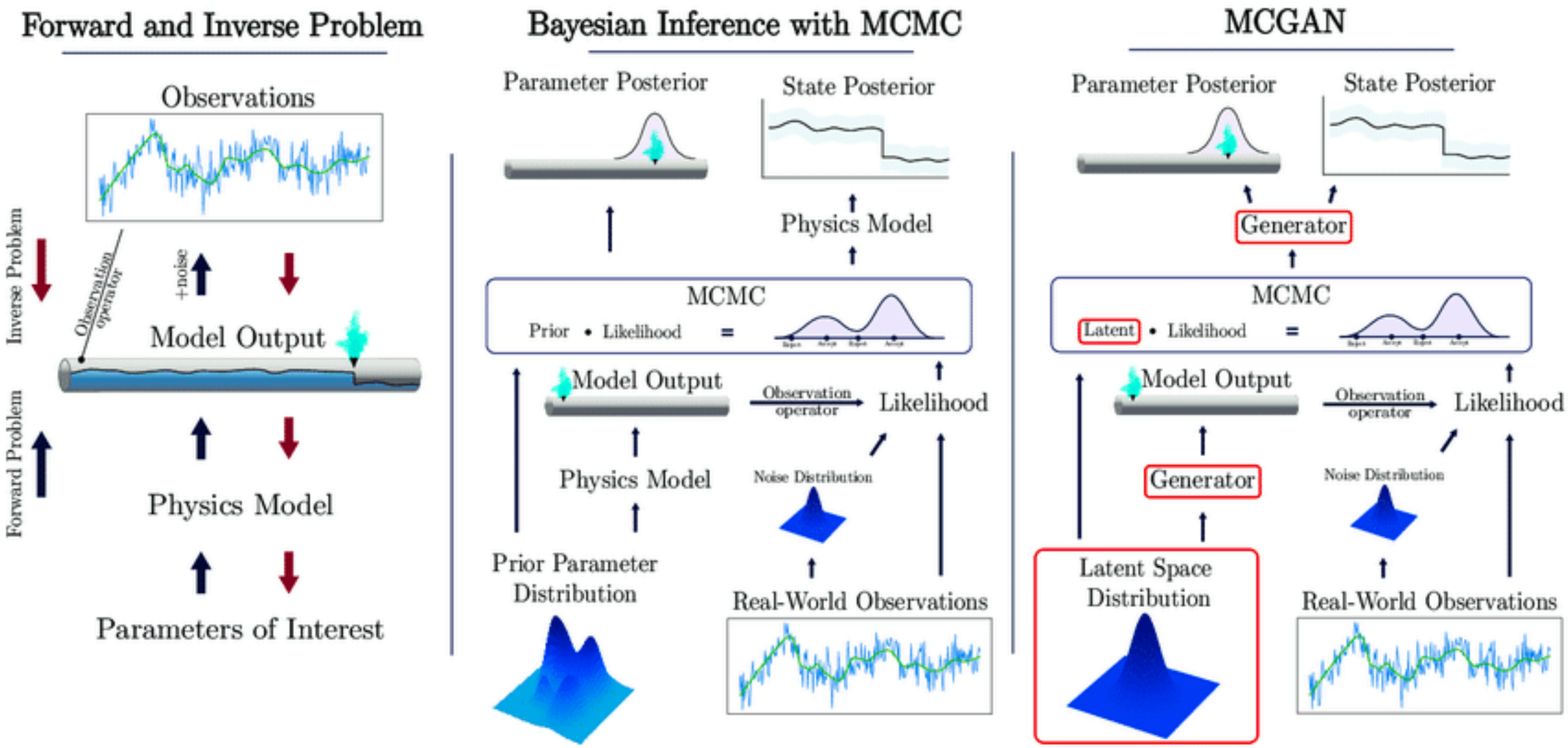
The inverse problem

Course: Physics of Complex Networks: Structure and Dynamics



Previously: Lab Session 3

(1) Simulate synchro dynamics for an ER and study how order parameter changes with coupling

(2) Change topology (WS, BA, SBM) and repeat

(3) Compare: which network synch first? (*Provided that other parameters are fairly equal*)

(4) Compare: Zachary network vs its configuration model (*over a set of realizations*)

Objective: gain (mis)information about structure or function

(Unknown) Equations of Dynamics

$$\dot{x}_i = f_i(x_1, x_2, \ldots, x_N, t) + \eta_i(t)$$

Correlation Matrix

$$C_{ij} = \rho(x_i(t), x_j(t))$$

Comparison

$$C_{ij} \text{ vs } A_{ij}$$

Weighted (statistical correlations) vs unweighted (structural) ??????

(1) Simulate white noises and build matrix C. Study the "topology" of C after applying thresholds (0.1, 0.2, …, 0.9) to cut off links less correlated than the threshold value. Use | Pearson |

(2) Same setup, but for each pair of time series devise a null model: simple reshuffling to destroy temporal correlations.

Calculate:

$$Z_{ij} = \frac{C_{ij}^{(data)} - \langle C_{ij}^{(reshuf)} \rangle}{\sigma_{ij}^{(reshuf)}} \qquad p_{ij} = 1 - \mathrm{erf}\left(\frac{Z_{ij}}{\sqrt{2}}\right)$$

(3) Same analysis, but using time series from the Kuramoto model

Workflow in general

(1) For each pair of signals (node i and j) calculate a correlation $C_{ij}^{(data)} = \rho\left[x_i^{(data)}(t), x_j^{(data)}(t)\right]$

(2) Build a surrogate signal according to a null model. For the null hypothesis of uncorrelated random noise):

$$\tilde{x}_i^{(1)}(t) = reshuff[x_i^{(data)}(t)]$$

(3) Calculate the correlation between surrogates: $C_{ij}^{(surr,1)} = \rho\left[\tilde{x}_i^{(1)}(t), \tilde{x}_j^{(1)}(t)\right]$

(4) Repeat points (2) and (3) for M times (M~100 is ideal), to obtain a set of correlation matrices

$$\left[C_{ij}^{(surr,1)}, \ldots, C_{ij}^{(surr,M)}\right]$$

(5) For each entry (i,j), calculate mean and standard deviation: $\langle C_{ij}^{(surr)}\rangle$, $\sigma_{ij}^{(surr)}$

(6) Estimate the Z-scores and the corresponding p-values (under the assumption that the underlying data is normally distributed):

$$Z_{ij} = \frac{C_{ij}^{(data)} - \langle C_{ij}^{(reshuf)}\rangle}{\sigma_{ij}^{(reshuf)}} \qquad p_{ij} = 1 - \mathrm{erf}\left(\frac{Z_{ij}}{\sqrt{2}}\right)$$

(7) Threshold and binarize: build a matrix B such that $B_{ij}^{(\alpha)} = 1$ if $p_{ij} \leq \alpha$ and $B_{ij}^{(\alpha)} = 0$ otherwise

(8) Assuming that B is the adjacency matrix of a network, study it for varying thresholds